

Subsurface Convection as the Origin of Stochastic Photometric Variability in Massive Stars

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Submitted to ApJ

ABSTRACT

Recent high-precision photometric observations have revealed a new ubiquitous phenomena in early type stars: stochastic low-frequency photometric variability. There are currently two main ideas for the origin of this photometric variability: Internal Gravity Waves (IGW) launched by the convective core, or granulation/IGW produced by (sub)surface convection. Thanks to high resolution ground-based spectroscopy, many of the targets observed with K2 and TESS now have also good stellar parameters, including mass, luminosity and a measure of surface velocity fluctuations (macroturbulence). Here we show theoretical predictions for the trend of characteristic frequency and amplitude of the perturbation induced by subsurface convection in massive stars, and compare these predictions to observations. We found that both the characteristic frequency ν_{char} and amplitude α_0 of the observed photometric variability correlate very well with the typical frequency τ_c^{-1} and flux of IGWs launched by the iron subsurface convection zone (FeCZ), respectively. While the amplitude of photometric variability also correlates with theoretical predictions of the flux of IGWs generated by core convection, the observed trend of characteristic frequency is at odds with theoretical expectations from convective turnover timescales in convective cores. This tentatively breaks the degeneracy between the two plausible models in favor of subsurface convection. We suggest a unified picture in which low-frequency photometric variability, surface turbulence, magnetic spots and wind variability in massive stars could all have the same origin: the presence of subsurface convection.

1. INTRODUCTION

Space Photometry revolution. New type of variability. Other mysteries at the surface of massive stars (turbulence, wind variability, spots). Stellar structure: convection in the core and convection in the envelope. IGWs and granulation. We discuss the properties of subsurface convection, in particular the maximum amplitude of waves that can be excited by their turbulent activity and their frequency as a function of the stellar temperature and luminosity. We show that the predicted trends of IGWs excited by turbulent convection in the FeCZ match very well the observations. We then proceed showing that while the trend with amplitude of the photometric variability could be reproduced by IGWs launched by core convection, the trend with excitation frequency in the core is opposite to the one observed at the surface. We discuss if wave propagation through

the stellar envelope can in principle revert this trend, and conclude this is unlikely. We conclude suggesting that subsurface convection represents a possible unifying mechanism for low-frequency photometric variability, surface turbulence, magnetic spots and wind variability in massive stars

This shows up as stochastic low-frequency variability that was observed in a large fraction of massive OB stars (see e.g. Balona 1992; Buysschaert et al. 2015; Bowman et al. 2019b).

1.1. Methods

We calculated stellar evolution models using the Modules for Experiments in Stellar Astrophysics (MESA Paxton et al. 2011, 2013, 2015, 2018, 2019) software instrument. Details on the microphysics inputs to this software instrument are given in Appendix A. The inlists used to run our models can be found in this [GitHub repository](#)¹. We focused on the structure and convective

properties of our models, in order to calculate typical frequencies and convective fluxes. Convection is calculated in the framework of MLT, and we adopted $\alpha_{\text{MLT}} = 1.6$. While in efficient convection zones (e.g. stellar cores) the choice of this parameter does not change results much, this is not true in superadiabatic, inefficient convective regions close to the stellar surface (Jiang et al. 2015; Cantiello & Braithwaite 2019). We discuss in Section 1.7 how uncertainties in the choice of the α_{MLT} parameter affect our results.

Since we are interested in the excitation of surface phenomena, we focus on the upper part of convective zones and following Cantiello et al. (2009) we define the average of a quantity $\bar{q}$ as

$$\bar{q} \equiv \frac{1}{\alpha_{\text{MLT}} H_{\text{P}}} \int_{R_{\text{c}} - \alpha_{\text{MLT}} H_{\text{P}}}^{R_{\text{c}}} q(r) dr \quad (1)$$

where H_{P} is the pressure scale height calculated at the upper boundary (R_{c}) of the convective zone of interest. We tested using different average procedures, and found that our results do not depend much on the specific choice.

Using this definition, we extracted the average Mach number $\bar{M}_{\text{c}}$, convective velocity $\bar{v}_{\text{c}}$ and density $\bar{\rho}_{\text{c}}$ in the convective core and in the subsurface convection zones of our models. The most important subsurface convection zone for the massive stars discussed in this work is the Iron Convection Zone (FeCZ) (Cantiello et al. 2009), although in the low luminosity regime He-driven convection zones could play a role as well (Cantiello & Braithwaite 2019). We define the characteristic frequency of convection as

$$\nu_{\text{char}} \equiv \frac{1}{2\pi\tau_{\text{c}}}, \quad (2)$$

where $\tau_{\text{c}} = \alpha_{\text{MLT}} \bar{H}_{\text{P}} / \bar{v}_{\text{c}}$ is the average convective turnover time in a convection zone. We proceeded to calculate convective turnover timescales and other relevant properties like convective fluxes.

We did this as a function of both mass and evolutionary history for stars ranging from 5 – 120 $M_{\odot}$, shown in Fig. ???. Here we present results for an initial metallicity of $Z=0.02$, but in Appendix B we report results for model grids with $Z = 0.014, 0.006$ and 0.002 as well.

1.2. Results

The presence of a subsurface convection zone can result in variability of photospheric properties. The simplified MLT treatment in our one dimensional calculations is unlikely to capture the complex phenomenology of these layers. Multi dimensional simulation including radiation have been performed in a restricted range of the parameter space, and show that the full turbulent

manifestation of these convective regions extends up to the stellar surface.

This can occur both because turbulent velocities excited by the convection extends to A proxy for the typical timescale of this variability is the convective turnover timescale. We found that the typical convective turnover timescale in the FeCZ is about $\sim \text{day}$, with a tendency for shorter values in models with high effective temperature and surface luminosity. To quantitatively compare to the characteristic frequencies of the observed stochastic photometric variability in massive stars, we proceed mapping the quantity ν_{char} (eq. 2) calculated in our models on the spectroscopic H-R diagram. This reveals a good agreement, with the models reproducing both the typical values of the characteristic frequency and the trend with $\log \mathcal{L}$ and $\log T_{\text{eff}}$ (Fig. 8).

1.3. Subsurface Convection and Macroturbulence

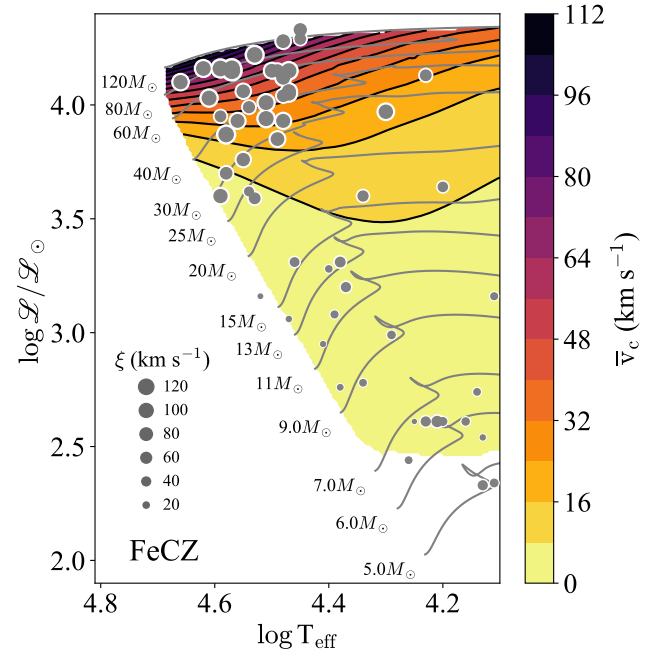


Figure 1. Average convective velocities in the FeCZ as function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected macroturbulent velocity as grey dots. The size of the dot is proportional to the observed macroturbulence.

1.4. Subsurface Convection and Photometric Variability

1.5. Core Convection and Photometric Variability

1.6. A unified model for surface phenomena in massive stars

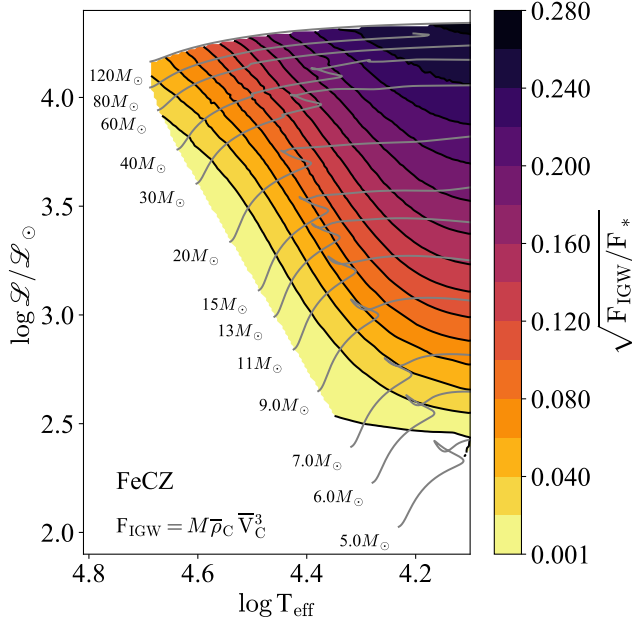


Figure 2. Relative flux of IGWs launched by the FeCZ. The IGWs flux is calculated multiplying the convective flux in the FeCZ by the average convective Mach number, calculated in the top pressure scale height of the convective region.

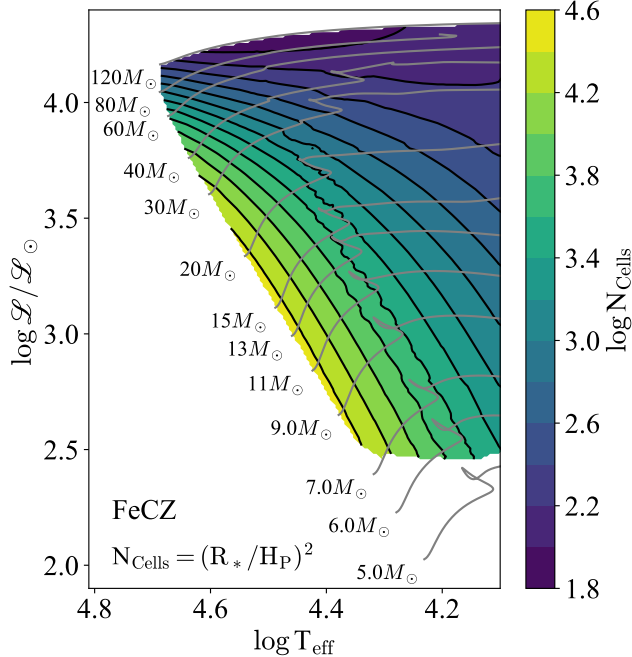


Figure 3. Expected number of convective cells in the FeCZ. This is calculated assuming the size of convective cells is about the average pressure scale height in the FeCZ

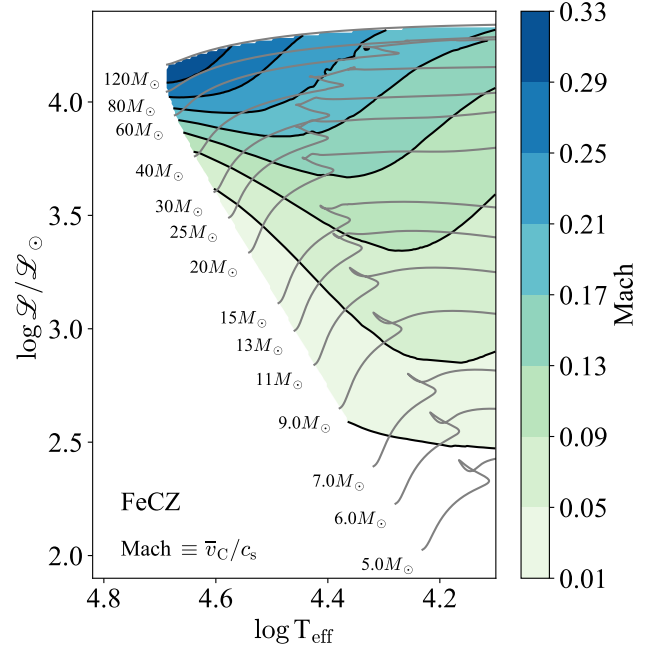


Figure 4. Mach number in the FeCZ.

The presence of subsurface convection can simultaneously account for a large variety of puzzling phenomena that appear ubiquitous at the surface of early-type stars. First of all we have microturbulence and macroturbulence. Both can be accounted for by velocity fields excited by the underlying FeCZ subsurface convection zone (Cantiello2009, Grassitelli 2015, This work). Photospheric magnetic spots: These spots might cause the ubiquitous DACs, resulting from CIRs in the winds of massive stars. The seed of wind clumping. Finally, the presence of a turbulent velocity field necessarily has to induce photometric variability, and we have shown here that the observed low-frequency stochastic noise observed in the majority of massive stars correlates very well with the presence and properties of the FeCZ. As such, a minimal hypothesis is that the presence of the FeCZ is the underlying physical cause and represent a possible unified model for the appearance of turbulence, magnetic spots, low-frequency photometric variability and wind structure and variability in massive stars.

1.7. Discussion

1.8. The Impact of α_{MLT}

1.9. Conclusions

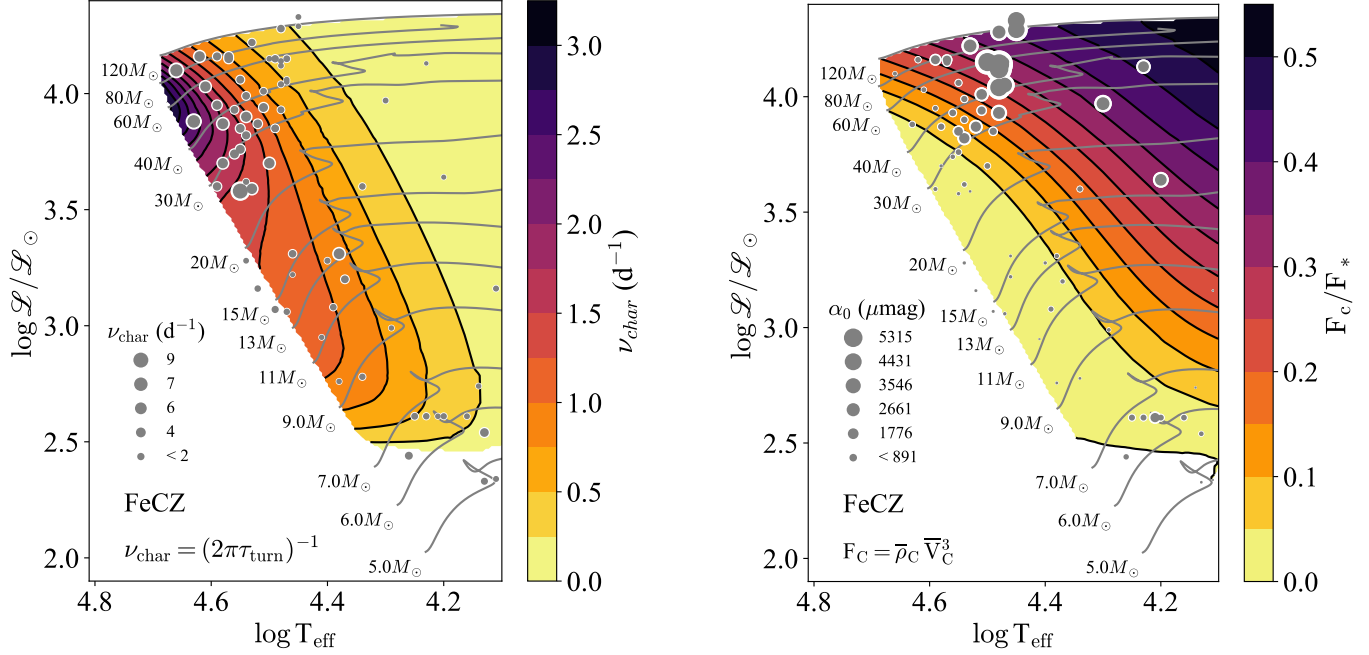


Figure 5. Left panel: Characteristic frequency ν_{char} in the FeCZ as a function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} . Right panel: Luminosity fluctuations expected due to IGWs launched by the FeCZ. This scales with the square root of the relative IGWs flux, with a correction factor accounting for the likely scale of the perturbation. We assumed uncorrelated perturbations at the surface with typical scale of a convective cell, so a correction factor $\sim 1/\sqrt{N_{\text{Cells}}}$. The IGWs flux is calculated multiplying the convective flux in the FeCZ by the average convective Mach number (this is calculated in the top pressure scale height of the convective region. The agreement is pretty good, considering that uncertainties of a factor of 2 in the MLT α parameters for these regions result in a factor of 20 in luminosity perturbations). We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

ACKNOWLEDGMENTS

The Center for Computational Astrophysics at the Flatiron Institute is supported by the Simons Foundation. This research was supported in part by the National Science Foundation under Grant No. NSF PHY-1748958 and by the Gordon and Betty Moore Foundation through Grant GBMF7392

APPENDIX

A. SOFTWARE DETAILS

Calculations were done with MESA version 11701. The MESA EOS is a blend of the OPAL Rogers & Nayfonov (2002), SCVH Saumon et al. (1995), PTEH Pols et al. (1995), HELM Timmes & Swesty (2000), and PC Potekhin & Chabrier (2010) EOSes.

Radiative opacities are primarily from OPAL (Iglesias & Rogers 1993, 1996), with low-temperature data from Ferguson et al. (2005) and the high-temperature, Compton-scattering dominated regime by Buchler & Yueh (1976). Electron conduction opacities are from Cassisi et al. (2007).

Nuclear reaction rates are a combination of rates from NACRE (Angulo et al. 1999), JINA REACLIB (Cyburt et al. 2010), plus additional tabulated weak reaction rates Fuller et al. (1985); Oda et al. (1994); Langanke & Martínez-

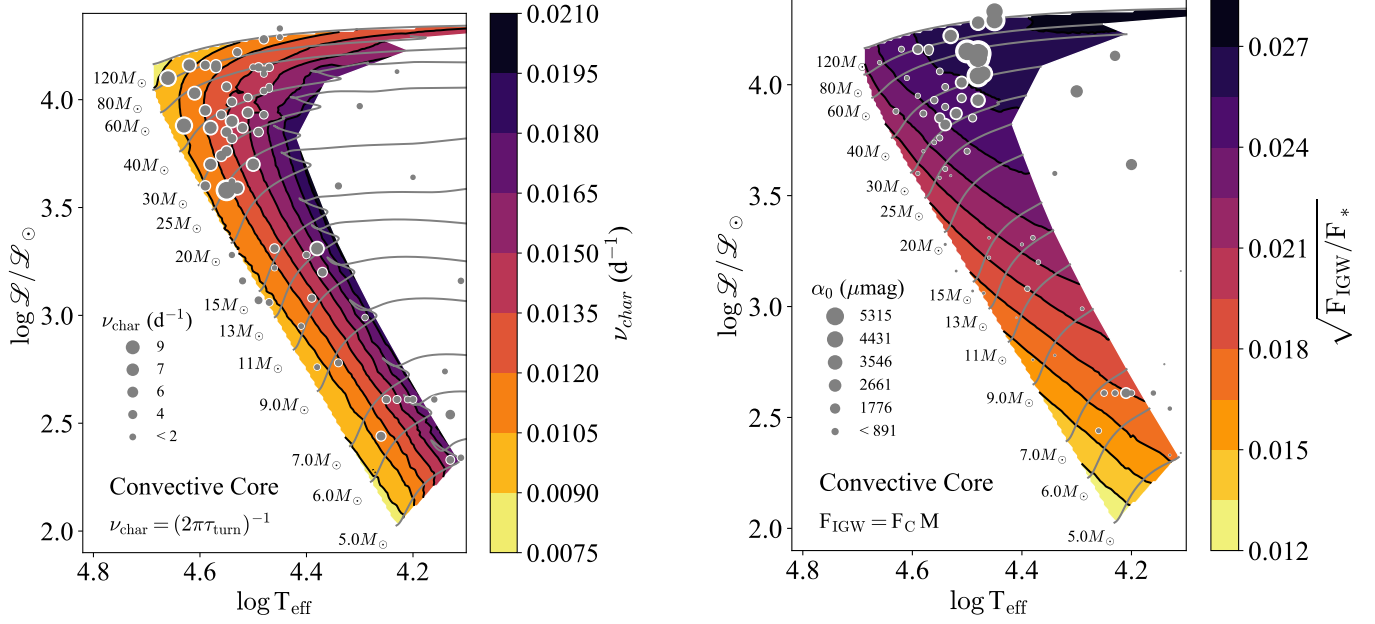


Figure 6. Left panel: Characteristic frequency ν_{char} in the convective core as function of the location of stellar models in the spectroscopic HRD. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} . Right panel: Flux of gravity waves launched by the convective core. This is calculated multiplying the convective flux in the core by the average convective Mach number (this is calculated in the top pressure scale height of the convective region). We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

Pinedo (2000). Screening is included via the prescription of Chugunov et al. (2007). Thermal neutrino loss rates are from Itoh et al. (1996).

The inlists, processing scripts, and model output are available at [Zenodo.org](https://zenodo.org).

B. GRIDS AT DIFFERENT METALLICITIES

Here we present results for the critical and surface magnetic field at metallicities of $Z=0.014$, $Z=0.006$, and $Z=0.002$, representing early-type stars in the Galaxy (MW), Large Magellanic Cloud (LMC), and Small Magellanic Cloud (SMC) respectively (e.g. Yusof et al. 2013). The initial Helium content of the grids is $Y=0.2659$ (MW), $Y=0.2559$ (LMC), and $Y=0.2508$ (SMC). The metallicity is initialized scaling the standard solar composition of Grevesse & Sauval (1998).

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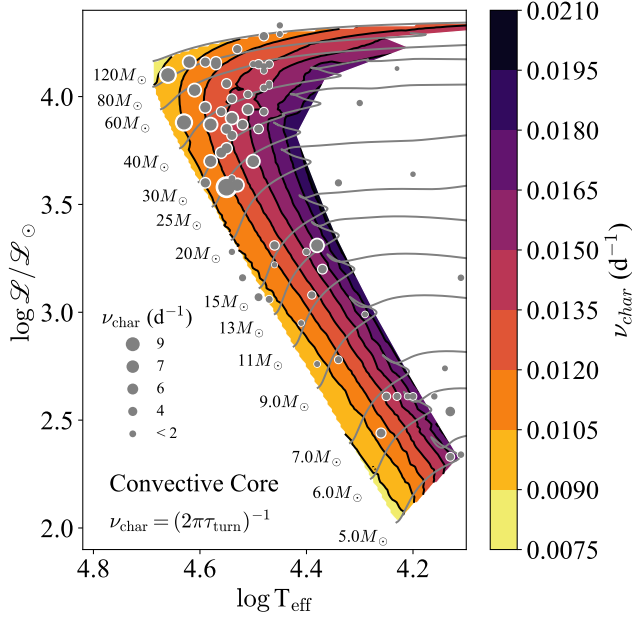


Figure 7. Characteristic frequency ν_{char} in the convective core as function of the location of stellar models in the spectroscopic HRD. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} .

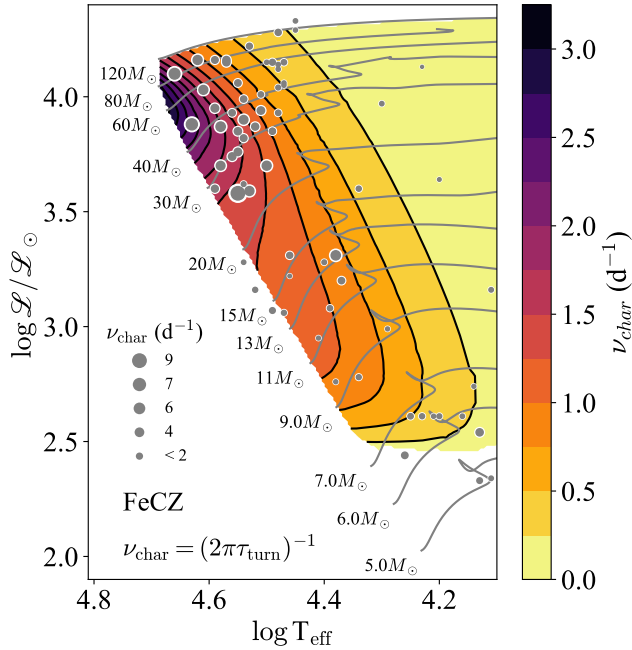


Figure 8. Characteristic frequency ν_{char} in the FeCZ as a function of the location of stellar models in the spectroscopic H-R Diagram. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to ν_{char} .

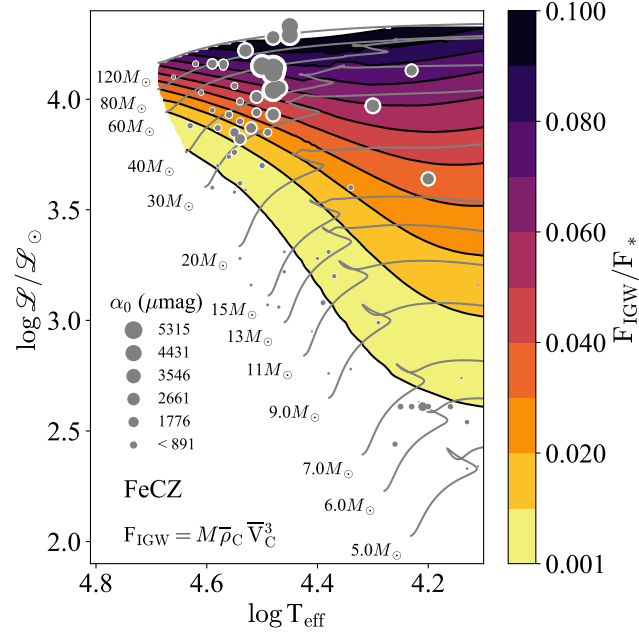


Figure 9. Flux of gravity waves launched by the FeCZ. This is calculated multiplying the convective flux in the FeCZ by the average convective Mach number (this is calculated in the top pressure scale height of the convective region). We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

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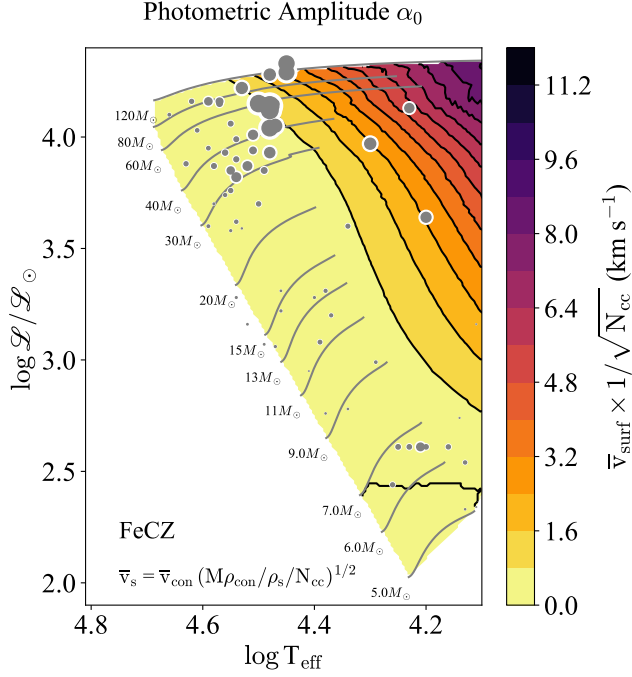


Figure 10. Reduced amplitude of velocity fluctuations at the stellar surface induced by internal gravity waves launched by the FeCZ in the spectroscopic H-R Diagram. After calculating the velocity fluctuation, we have corrected for cancellation effects dividing by the total number of convective cells. We also show the observed stars with detected stochastic photometric variability as grey dots. The size of the dot is proportional to α_0 .

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